

Python Student Progress & Performance Tracking System

A Centralised School Management System

By Group 2

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Purpose

01

Centralised
Academic
Management
System

02

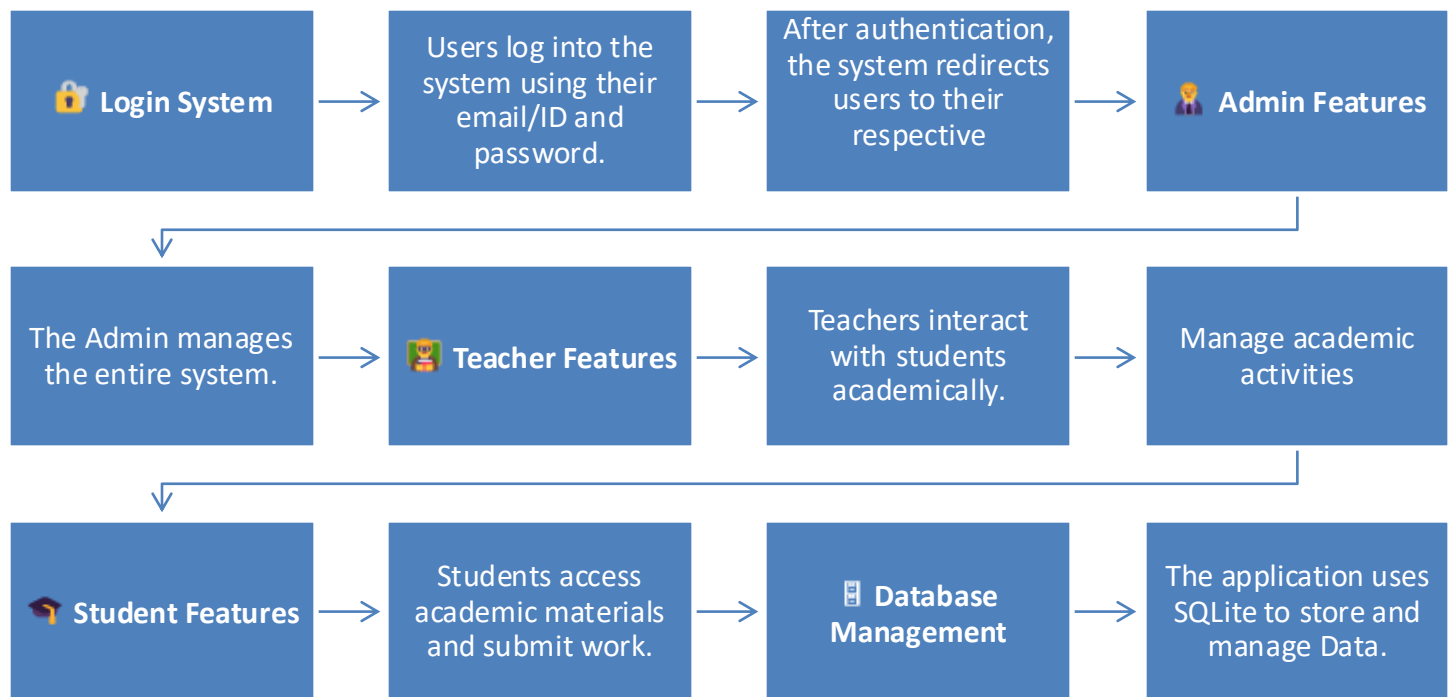
Manage
students,
teachers,
assignments

03

Automate
submissions
and grading



How the Application Works



System Roles



ADMIN: MANAGES
USERS



TEACHER: CREATES
ASSIGNMENTS, MARKS



STUDENT: SUBMITS
WORK, VIEWS RESULTS



What This Application Replaces

- Before this application, many academic processes were handled manually using:
- Paper records
- Physical assignment submissions
- Manual grading sheets
- Spreadsheet tracking
- Verbal communication
- This application replaces those manual systems with a centralised digital platform.



Benefits of the Application

- Faster academic management
- Centralised student records
- Easier assignment tracking
- Secure user authentication
- Improved communication
- Reduced paperwork
- Better organisation and efficiency

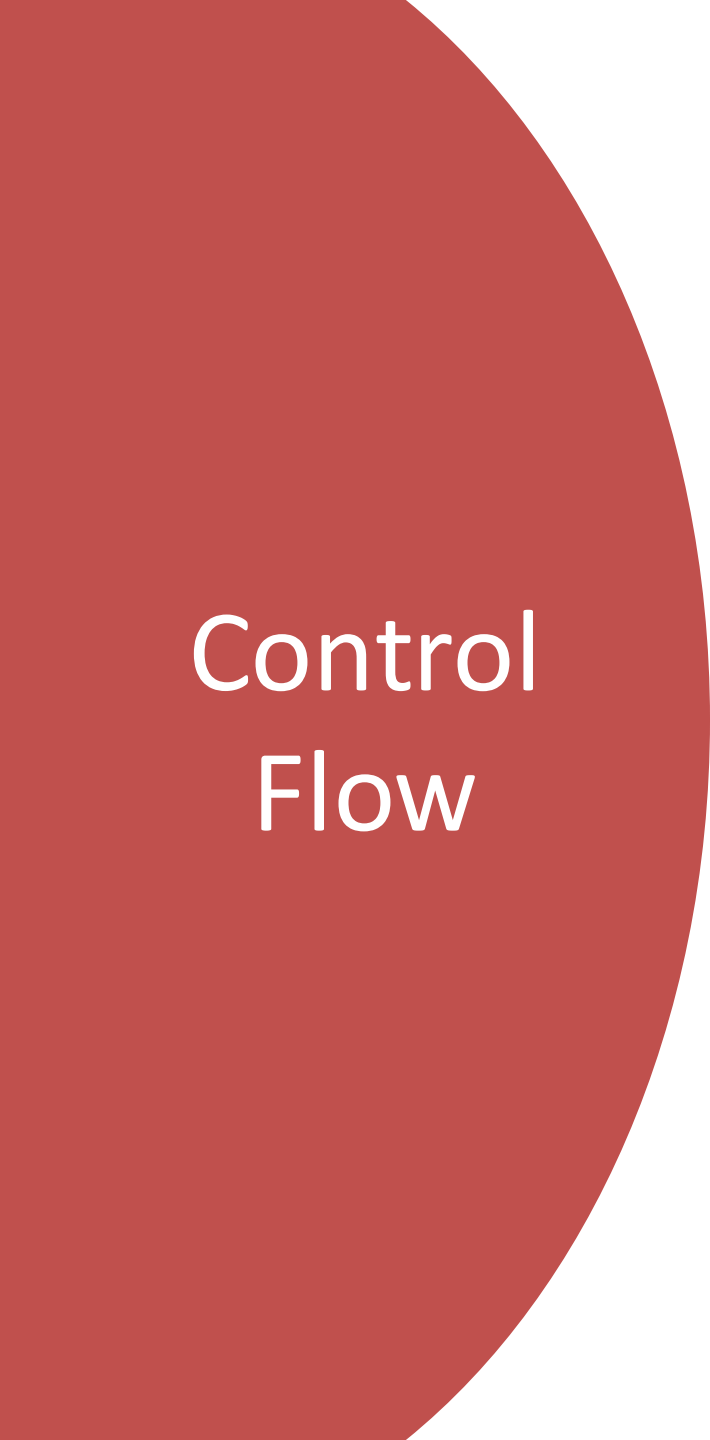


System Architecture

UI Layer: PyQt6

Logic Layer: managers
(auth, assignments)

Database Layer:
SQLite



Control Flow

Login → Role detection
→ Dashboard

Teacher: Assign →
Grade

Student: Submit →
View marks

System Diagram

Login UI



Auth System



Admin / Teacher / Student



Database Operations

Step-by-Step Guide to Building the Python Management Application



Planning Phase

- Define the purpose of the App
- Identify user roles:
 - Admin
 - Teacher
 - Student

List required features:

- Login system
- Assignment management
- Submissions
- Marks
- Notifications
- Design basic workflow and database structure

Set up, Create

- **Database Setup**

- Create SQLite database
- Create tables:
 - users
 - students
 - teachers
 - assignments
 - submissions
 - marks
 - notifications
- Add primary and foreign keys

- **Authentication System**

- Build login functionality
- Hash passwords using SHA-256
- Redirect users based on role

- **GUI Development (PyQt6)**

Create:

Login window
Admin dashboard
Teacher dashboard
Student dashboard

Use:

Buttons
Tables
Forms
Layouts
Message boxes

- **Admin Module**

Implement:

Add students
Add teachers
View records
Manage users

- **Teacher Module**

Implement:

Create assignments
Upload assignment files
View submissions
Add marks and feedback

- **Student Module**

Implement:

View assignments
Download assignment files
Upload submissions
View marks and notifications

More Set up & Finalising

- **UI Improvement**

- Add:
- Colors
- Fonts
- Button styling
- Dashboard layouts
- Better spacing and alignment
-

- **Documentation**

- Prepare:
- README file
- Presentation slides
- System diagrams
- User guide
-

- **Final Deployment**

- Run application using main.py
- Ensure database and upload folders exist
- Perform final testing before presentation

- **File Handling**

Use:

`os.makedirs()` for folders

`shutil.copy()` for uploads

Create:

uploads/assignments

uploads/submissions

- **Testing & Debugging**

Test:

Login system

Assignment uploads

Submission process

Foreign key relationships

GUI navigation

Fix:

Import errors

Database constraints

File path issues



Project File Structure

App_Project/

|

└─ main.py

└─ database.py

└─ auth.py

└─ login_window.py

|

└─ admin_dashboard.py

└─ teacher_dashboard.py

└─ student_dashboard.py

|

└─ student_manager.py

└─ teacher_manager.py

└─ assignment_manager.py

└─ submission_manager.py

└─ marks_manager.py

└─ notifications.py

└─ reports.py

|

└─ uploads/

| └─ assignments/

| └─ submissions/

|

└─ App.db

Tools Used

Python

PyQt6 (GUI)

SQLite
(Database)

Hashlib
(Security)

OS & Shutil
(File handling)

Challenges



FOREIGN KEY
ERRORS



LOGIN ISSUES



FILE UPLOAD
HANDLING



GUI LAYOUT
PROBLEMS



Challenges



Additional Project Challenges



Coordinating a group of members effectively



Communication gaps within the team



Delays caused by personal and academic responsibilities



Maintaining motivation and consistent contribution from all members



Managing time and balancing workloads across the team

Future Improvements



Cloud
database



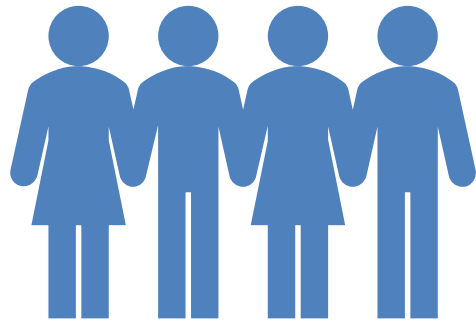
Better UI
design



Mobile
version



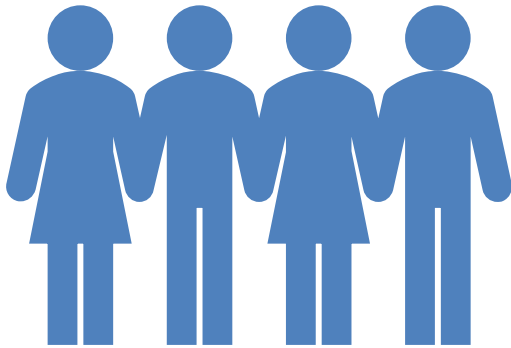
Real-time
notifications



Lessons Learned During Development

- Team communication is critical in group projects
- Planning ahead reduces delays and technical issues
- Clear role allocation improves productivity
- Leadership and motivation are important for team success
- Testing frequently helps identify issues early
- Anticipating challenges early helps reduce project delays
- Keeping all team members aligned improves productivity
- Building an App is interesting. It gives you a good sense of purpose

**Thank you for giving
us your time**



The end